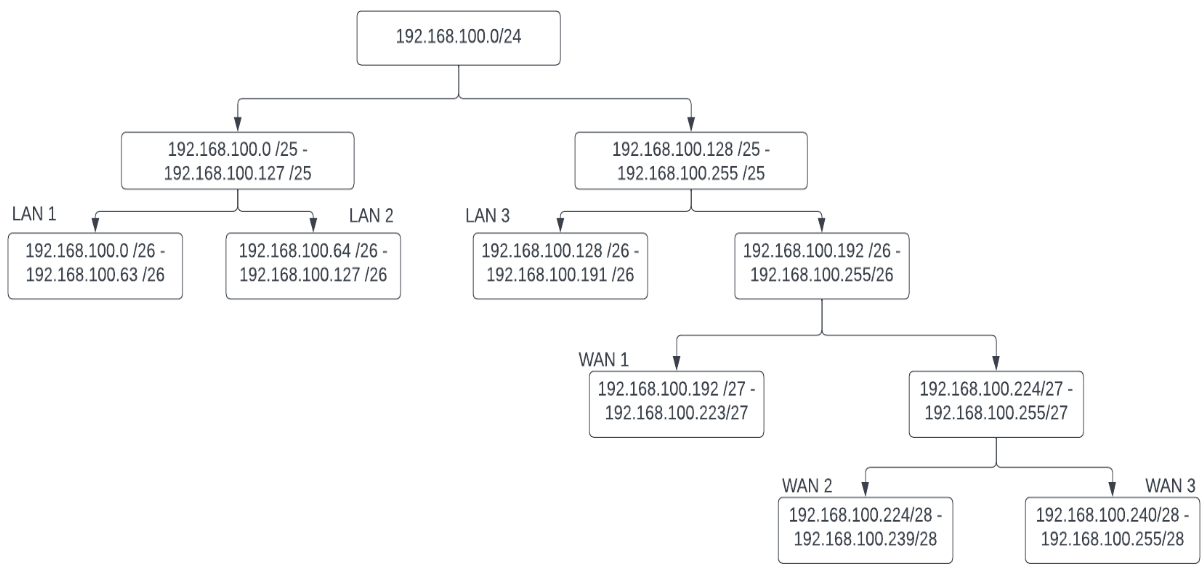
Networking Technologies

Step 1: Determining the base address

First of all, we need to determine the network base address. According to the given instructions, the network address would be 192.168.100.0/24.

Step 2: Subnetting the Base Network

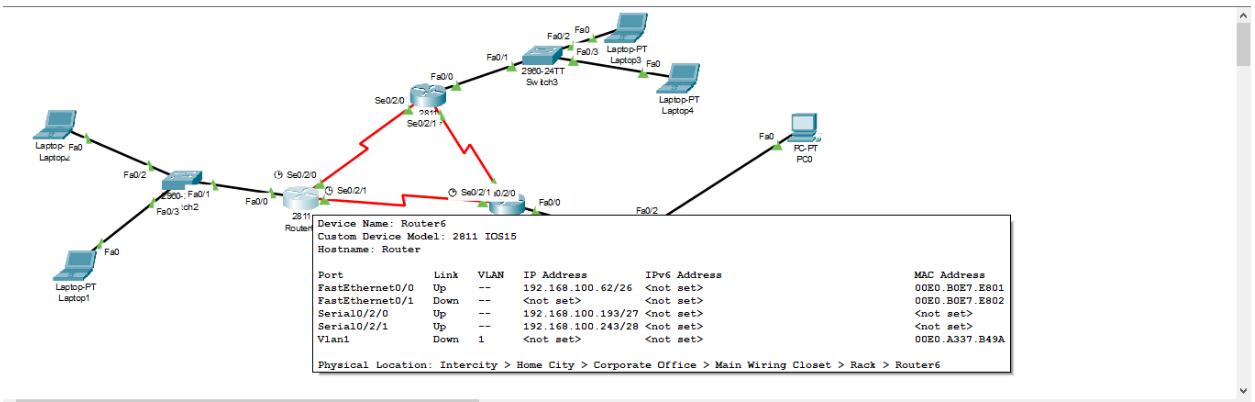


LAN 1 Subnet:
To create the network address for LAN 1, I have borrowed 6 bits from original base network i.e 192.168.100.0/24.

- Borrowed Bits: 6 (from the original 24 bits)
- New Subnet Mask: 255.255.255.192 (because 26 bits are used for network portion)
- IP Range for LAN1: 192.168.100.1 to 192.168.100.62
- Usable IP Addresses for LAN1: 192.168.100.1 to 192.168.100.62
- Subnet Broadcast Address: 192.168.100.63 (last address in the subnet)
- Network assigned to switch: 192.168.100.62

- Network assigned to PC1: 192.168.100.1
- Network assigned to PC2: 192.168.100.2

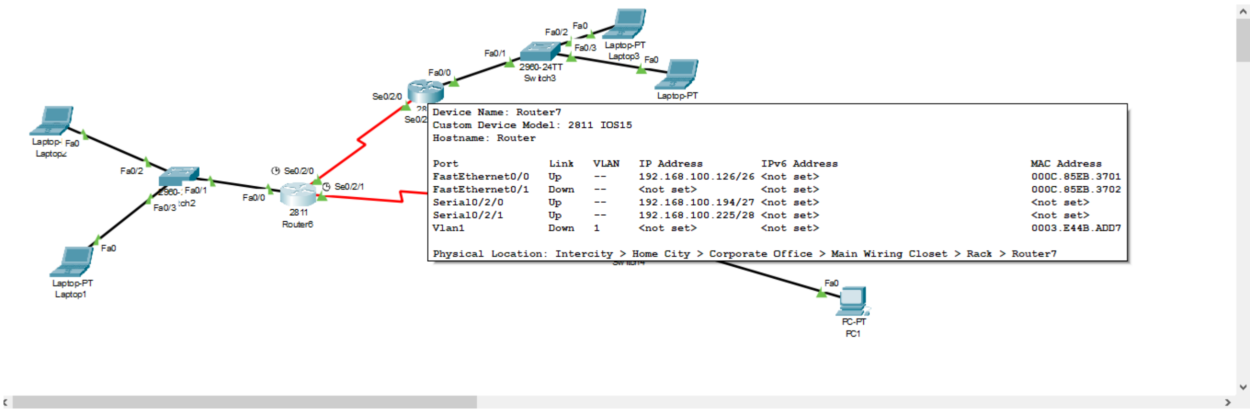
So, for LAN1, I have a subnet mask of 255.255.255.192, which means 192.168.100.0 to 192.168.100.63 is my address range. 192.168.100.0 is the network address, 192.168.100.63 is the broadcast address, and 192.168.100.1 to 192.168.100.62 are usable IP addresses for devices within this subnet.



LAN 2 Subnet:
To create the network address for LAN 2, I have borrowed 6 bits from original base network i.e 192.168.100.0/24.

- Borrowed Bits: 6 (from the original 24 bits)
- New Subnet Mask: 255.255.255.192 (because 26 bits are used for the network portion)
- IP Range for LAN2: 192.168.100.64 to 192.168.100.127
- Usable IP Addresses for LAN2: 192.168.100.65 to 192.168.100.126
- Subnet Broadcast Address: 192.168.100.127 (last address in the subnet)
- Network assigned to switch: 192.168.100.126
- Network assigned to PC1: 192.168.100.65
- Network assigned to PC2: 192.168.100.66

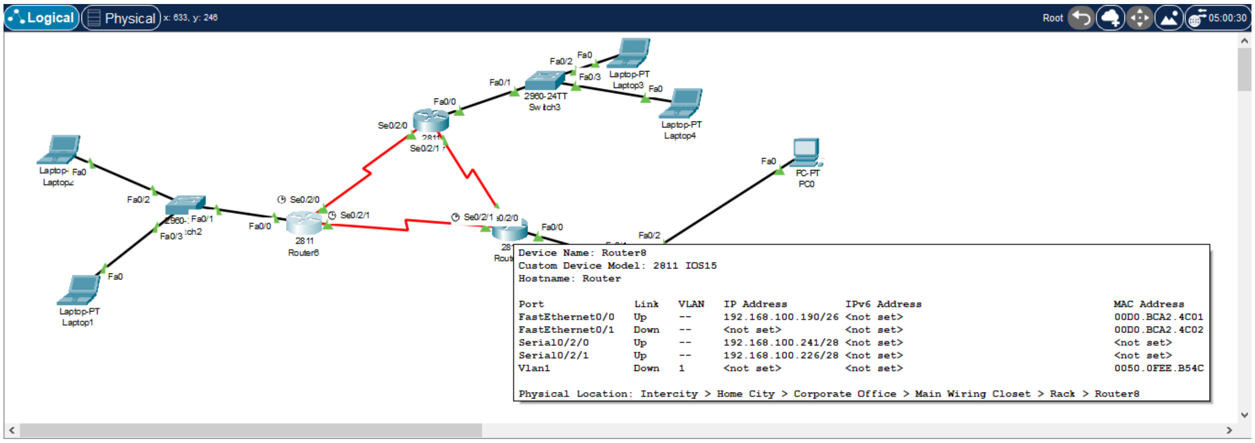
So, for LAN2, I have a subnet mask of 255.255.255.192, which means 192.168.100.64 to 192.168.100.127 is my address range. 192.168.100.64 is the network address, 192.168.100.127 is the broadcast address, and 192.168.100.65 to 192.168.100.126 are usable IP addresses for devices within this subnet.



LAN 3 Subnet:
To create the network address for LAN 3, I have borrowed 6 bits from original base network i.e 192.168.100.0/24.

- Borrowed Bits: 6 (from the original 24 bits)
- New Subnet Mask: 255.255.255.192 (because 26 bits are used for the network portion)
- IP Range for LAN3: 192.168.100.128 to 192.168.100.191
- Usable IP Addresses for LAN3: 192.168.100.129 to 192.168.100.190
- Subnet Broadcast Address: 192.168.100.191 (last address in the subnet)
- Network assigned to switch: 192.168.100.190
- Network assigned to PC1: 192.168.100.129
- Network assigned to PC2: 192.168.100.130

So, for LAN3, I have a subnet mask of 255.255.255.192, which means 192.168.100.128 to 192.168.100.191 is my address range. 192.168.100.128 is the network address, 192.168.100.191 is the broadcast address, and 192.168.100.129 to 192.168.100.158 are usable IP addresses for devices within this subnet.



WAN 1 Subnet:
In WAN 1, I have assigned IP addresses 192.168.100.193/27 to a SE0/2/0 port on Router 1 connecting to Router 2 and IP address 192.168.100.194/27 to a SE0/2/0 port on Router 2.

- Subnet Mask: 255.255.255.224
- Network Address: 192.168.100.192
- Broadcast Address: 192.168.100.223
- Usable IP Address Range: 192.168.100.193 to 192.168.100.222

Both routers on WAN 1 are configured with IP addresses in this subnet, and the address range allows for two usable IP addresses (192.168.100.193 and 192.168.100.194) for the point-to-point link.

WAN 2 Subnet:
In WAN 2, I have assigned IP addresses 192.168.100.225/28 to a SE0/2/1 port on Router 2 connecting to Router 3 and IP address 192.168.100.226/28 to a SE0/2/1 port on Router 3.

- Subnet Mask: 255.255.255.240
- Network Address: 192.168.100.224
- Broadcast Address: 192.168.100.239
- Usable IP Address Range: 192.168.100.225 to 192.168.100.238

Both routers on WAN 2 are configured with IP addresses in this subnet, and the address range allows for two usable IP addresses (192.168.100.225 and 192.168.100.226) for the point-to-point link.

WAN 3 Subnet:

In WAN 3, I have assigned IP addresses 192.168.100.241/28 to a SE0/2/0 port on Router 3 connecting to Router 1 and IP address 192.168.100.243/28 to a SE0/2/1 port on Router 1.

- Subnet Mask: 255.255.255.240
- Network Address: 192.168.100.240
- Broadcast Address: 192.168.100.255
- Usable IP Address Range: 192.168.100.241 to 192.168.100.254

Both routers on WAN 3 are configured with IP addresses in this subnet, and the address range allows for two usable IP addresses (192.168.100.241 and 192.168.100.243) for the point-to-point link.

Role of RIP as a Routing Protocol

Routing Information Protocol (RIP) is a distance-vector routing protocol that plays a crucial role in the exchange of routing information within a network.

Role of RIPv2 as a Routing Protocol:

Routing protocols like RIPv2 are essential for dynamically sharing routing information among routers within an internetwork. RIPv2 offers the following key roles and functionalities.

- **Routing Information Exchange:**
  RIPv2 allows routers to exchange routing information, including network addresses and routing metrics, which helps them build and maintain their routing tables. This exchange enables routers to make informed decisions about the best path to reach a destination network.

- **Convergence:**
  RIPv2 facilitates network convergence by quickly adapting to changes in the network topology. When a network link or router fails

or when new routes become available, RIPv2 routers exchange updates to reflect these changes, allowing for timely adaptation.

- **Distance-Vector Algorithm:**
  RIPv2 employs a distance-vector routing algorithm. It calculates routing metrics based on the number of hops or "distance" to a destination network. The goal is to choose the path with the fewest hops, making it suitable for small to medium-sized networks.

- **Ease of Configuration:**
  RIPv2 is relatively straightforward to configure, making it accessible for network administrators. It is particularly well-suited for simple, homogeneous network topologies.

- **Subnet Support**:
  One significant improvement in RIPv2 over RIPv1 is its support for subnetting. RIPv2 can carry subnet mask information within its routing updates, allowing for more precise routing decisions in modern, subnetted networks.

Comparison of RIPv1 and RIPv2:

RIPv1, the earlier version of RIP, had limitations that were addressed in RIPv2. The primary differences between the two versions are:

- **Classless Routing:**
  RIPv1 assumes classful addressing, which does not support Variable Length Subnet Masks (VLSM). RIPv2, on the other hand, is classless and supports subnetting, making it compatible with modern, more flexible network designs.

- **Authentication:**
  RIPv1 lacks authentication mechanisms, making it vulnerable to unauthorized or malicious routing updates. RIPv2 offers built-in authentication for secure routing information exchange.

- **Route Tagging:**
  RIPv2 supports route tagging, which allows routers to identify routes that have been redistributed from other routing protocols. This feature is useful when integrating multiple routing protocols within the same network.

Use of RIPv2 in provided scenario:

The scenario presented involves advertising sub netted network addresses, which immediately necessitates the use of RIPv2 over RIPv1. In a network where subnetting is essential for efficient address allocation and routing, RIPv2 is the more suitable choice due to the following reasons:

- **Subnet Support:**
  RIPv2 can carry subnet mask information within its routing updates. This means it can provide detailed information about subnets within a network, enabling routers to make more precise routing decisions. This is especially important when subnetting is used to efficiently manage IP address space.
- **Classless Routing:**
  RIPv2 is classless, which means it can work with networks that use VLSM. In modern networks, it is common to use VLSM to optimize address allocation and network management. RIPv2's support for classless routing is a significant advantage in such scenarios.
- **Authentication:**
  Security is a critical concern in modern networking. RIPv2 provides built-in authentication, ensuring that routing updates are only accepted from authorized routers. This is crucial to prevent unauthorized or malicious changes to the routing tables.

Configuration of RIPv2 in Cisco IOS:

To configure routers to employ RIPv2 for advertising sub netted network addresses in Cisco IOS, the following steps are typically performed:

1. Access the router's **command-line interface (CLI)** through a terminal emulation software.
2. Enter global configuration mode by typing **configure terminal or conf t**.
3. Enable the RIP routing process by typing **router rip**.

4. Specify the version as RIPv2 by using the **version 2** command.
5. Add network statements to specify which interfaces are part of the RIP process. For example, **network 192.168.1.0** would include the 192.168.1.0 network in the RIP updates.
6. Save the configuration with the **copy running-config startup-config** command.

In conclusion, RIPv2 is a valuable routing protocol for sub netted networks due to its support for classless routing, authentication, reduced network overhead, and route tagging. Configuring RIPv2 in Cisco IOS involves enabling the protocol, specifying the version, adding network statements, and optionally configuring authentication. When deploying sub netted networks and maintaining security, RIPv2 is the preferred choice over RIPv1.

Configuring Routers and Switches:

Step 1: Initial Device Configuration

- **Access the device:** Enter the device's command-line interface through the console connection
- **Enter privileged exec mode:** Type enable and provide the enable secret or enable password to access privileged exec mode.

Step 2: Secure Access to the Device

- **Create a secure password:** Set an enable secret password using the following command:

  enable secret <password>

- **Implement SSH:** Disable Telnet and enable Secure Shell (SSH) for remote management. Use the following commands:

  line vty 0 4

  transport input ssh

  login local

Step 3: Configuring Router Interfaces

- Access the interface configuration mode for each interface you want to configure.

  interface FastEthernet0/0
- Configure the IP address and subnet mask for each interface using the ip address command:

  ip address <ip_address> <subnet_mask>
- Use the following command to activate the interface.

  No shutdown

Step 4: Configure Routing Protocol (RIPv2)

- Enter global configuration mode and activate the RIPv2 routing process.

  router rip

  version 2
- Specify the networks to advertise in RIP updates:

  network <network_address>

Step 5: Set Password on Switches:

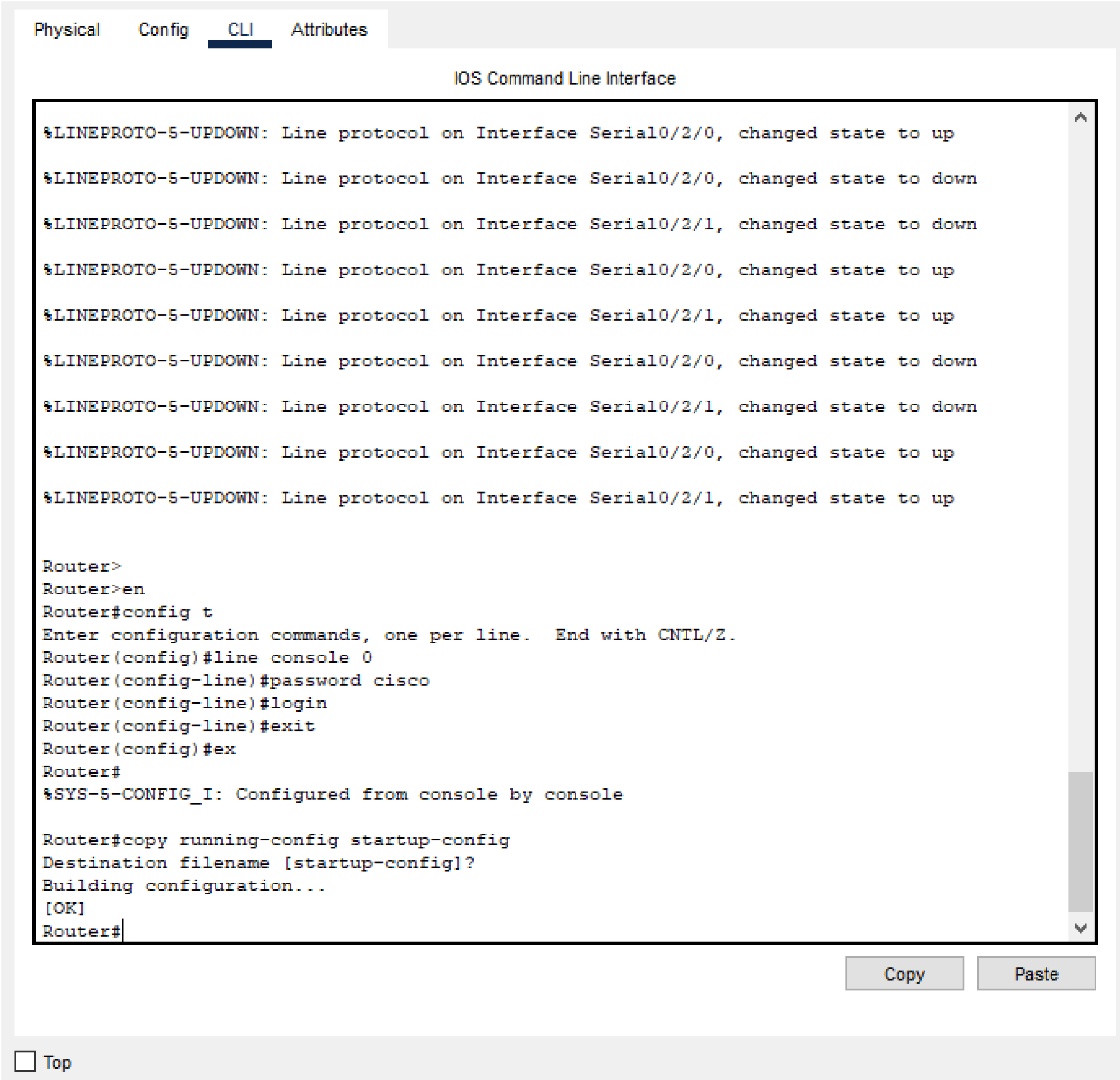
Set up a password on switches by following command:

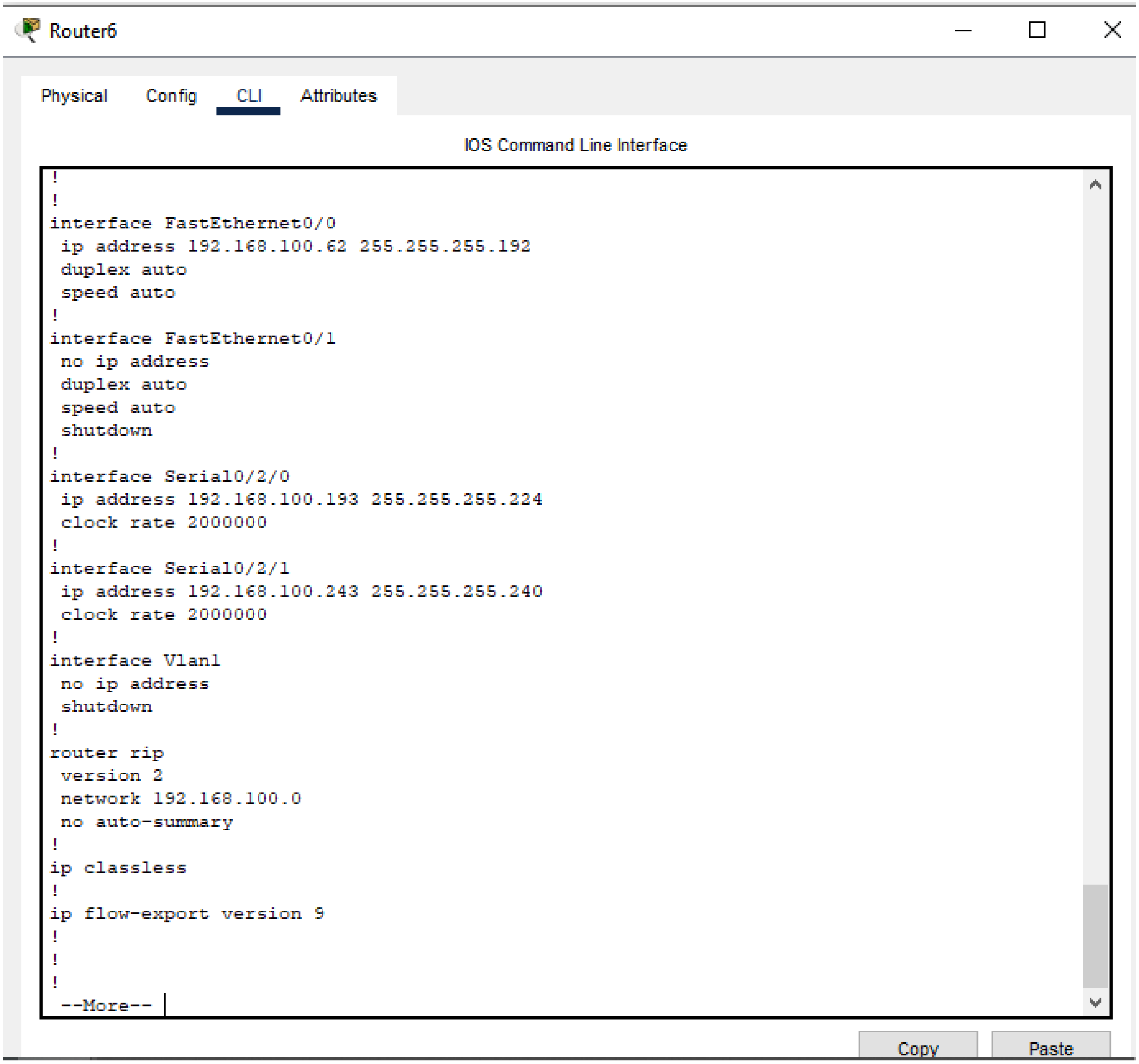
enable secret <password>

Step 6: Save Configuration

After completing the configuration, ensure to save the changes to the startup-config file to make them persistent. Use **copy running-config startup-config** command.

Above mentioned steps were used to configure your routers and switches in Packet Tracer to implement your network design while enhancing security against unauthorized configuration changes. Following are the screenshots attached for the above-mentioned steps:





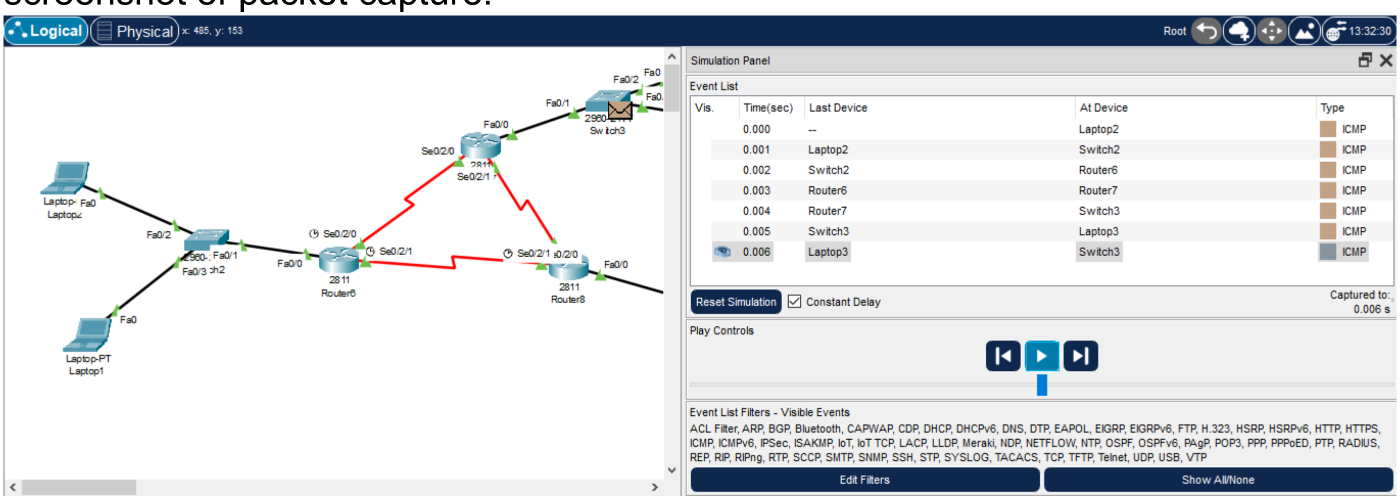
The above screenshots show the security enabled on a router and RIP configuration is used in a router. Similarly, it has been implemented on other routers too.

Troubleshooting Tools:

1. Packet Capture:

In Cisco Packet Tracer, packet capture is a valuable troubleshooting tool that allows you to capture and analyze network traffic within the simulated environment. Packet capture is often used to diagnose network issues, monitor communication between devices, and verify the correctness of network configurations.

Working:
To use it, open your network topology, select the device or interface you want to capture packets on (like a router or switch), right-click, and enable packet capture. This starts recording incoming and outgoing network packets. You can then view and analyze these packets, inspecting details like source and destination IP addresses and port numbers. You can also filter or search for specific data within the captured packets. To stop the capture, return to the right-click menu and disable it. This captured data is invaluable for troubleshooting network issues, as it helps identify connectivity problems, communication patterns, errors, and configuration errors. After analysis, you can take informed actions to resolve problems or enhance network performance. You can also save and export captured packets for further investigation, making it a powerful tool for understanding network behavior within a simulated environment. Following is attached screenshot of packet capture:

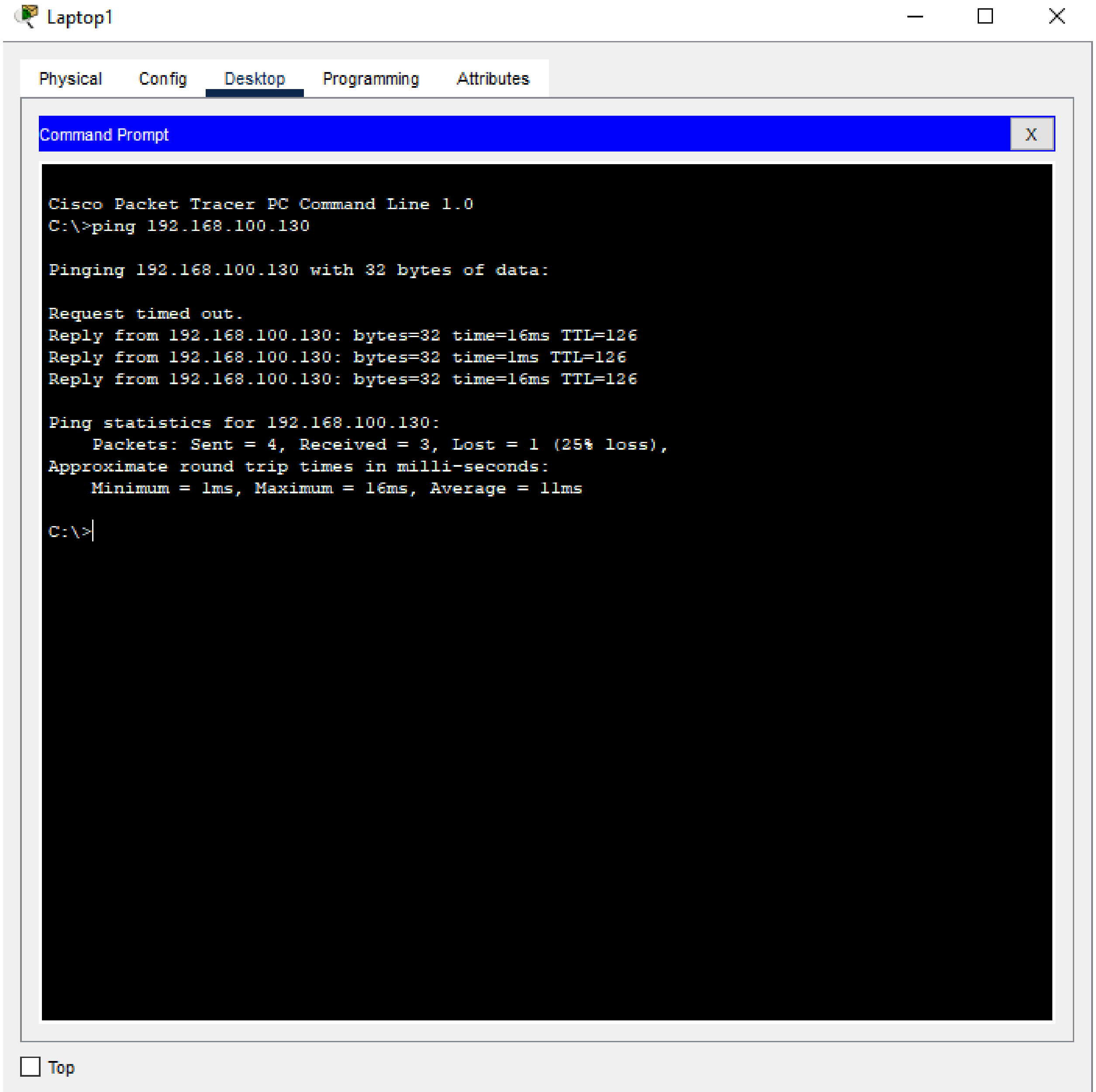


2. Ping:

Ping is a versatile troubleshooting tool in Cisco Packet Tracer. It helps in various aspects of network diagnostics. Firstly, it checks connectivity by sending packets to a destination; receiving replies indicates the device is online, while no replies may suggest network issues or an offline destination. Additionally, Ping measures network latency, with a longer round-trip time (RTT) indicating potential network congestion or connection problems. It's also effective for detecting packet loss, which can signal network problems like congestion or misconfigured devices. Furthermore, Ping can be used to verify the path packets take to their destination, revealing routing issues if packets follow unexpected routes. These capabilities make Ping an essential tool for assessing and troubleshooting network performance and connectivity.

Working:

To utilize Ping in Cisco Packet Tracer, you initiate it from the command-line interface (CLI) on a router, switch, or end device of your choice. Within the CLI, you enter the command: "ping <destination_IP_address>," with "<destination_IP_address>" being replaced by the specific IP address of the device you intend to ping. Upon executing this command, your device sends out a sequence of ICMP (Internet Control Message Protocol) Echo Request packets directed to the provided destination IP address. If the destination device is indeed reachable and responsive, it will promptly respond with ICMP Echo Reply packets. Through this exchange, Ping measures the round-trip time (RTT), expressed in milliseconds, showcasing the duration it takes for a packet to journey from the sender to the destination and back. This real-time feedback is invaluable for assessing network connectivity and diagnosing potential issues.



3. Show Commands:

Show commands in Cisco Packet Tracer are a set of diagnostic commands used for troubleshooting and monitoring the status and configuration of network devices, such as routers and switches. These commands provide real-time information about various aspects of a network, helping network administrators identify and address issues.

Working:

Using show commands in Cisco Packet Tracer begins with accessing the device's command-line interface (CLI). Once inside the CLI, you input show commands with a specific keyword to retrieve information. For example, "show ip interface brief" displays a summary of IP interfaces. The device then provides data related to the command's purpose, such as configuration details, interface statuses, routing tables, or ARP tables. Analyzing the output is crucial, looking for errors or inconsistencies that might point to network issues. Interpreting the results allows you to identify potential problems. For instance, missing or incorrect routes revealed by "show ip route" can explain connectivity issues. Once an issue is identified, you can take corrective actions, like modifying configurations or addressing errors to resolve network problems. Show commands are valuable for both troubleshooting and ongoing network monitoring, making them essential tools for network administrators.